Introduction:Digital library

Outline

- Digital Library and E-learning
  - Vision
  - Definition
  - Theoretical framework
- Digital Library as Learning Environment
- Learning Object
- Digital Librarians as Educators

Reordering Ranganathan: Shifting User Behaviors, Shifting Priorities

Connaway L. 2014

	Ranganathan's Original Conception	New Conceptions in the Current Environment		
First Law	Books are for use.	E-books are for reading.	Netflix is for watching.	Blackboard is for studying.
Second Law	Every person his or her book.	Every listener her iTunes.	Every artist his Photoshop.	Every student her EasyBib.
Third Law	Every book its reader.	Every blog its reader.	Every Google Map, its traveler.	Every digital repository its researcher.
Fourth Law	Save the time of the reader.	Save the time of the listener.	Save the time of the traveler.	Save the time of the researcher.
Fifth Law	A library is a growing organism.			

- Digital libraries have become the inevitable part of the contemporary information society for storage and access of huge amount of information. Easy access to this vast treasure of knowledge is possible through bit stream or digital content. The DLs inherent advantages like high volume of storage of multiform information, effective search and retrieval, accessible through multiple points at any time, instantaneous downloading, faster addition of information etc. made the Government to initiate digital library projects.

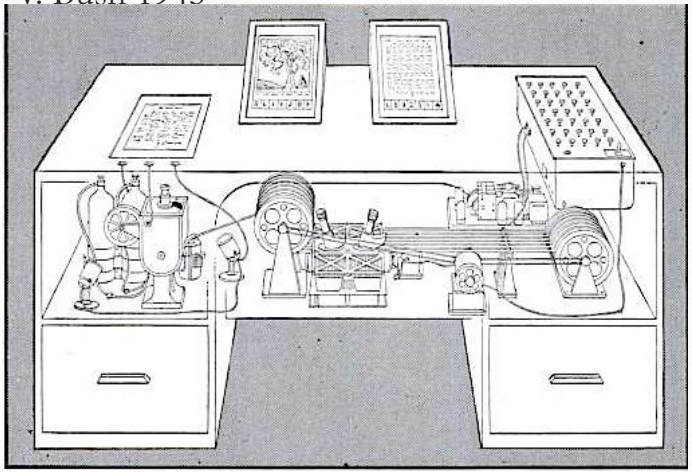
- Ian Witten et al identified four major roles of digital libraries.
- 1. Support human development: With increase of information resources on CD/DVD made available in rural / remote areas of developing countries; their cost and maintenance is much lesser than the paper books digital libraries support human development. Ex. UN documents on agriculture, environment, Medicine, Health etc. reaching to rural areas in Uganda
- 2. Pushing on the frontiers of Science: Through archival digital libraries that are developed with the contributions of scientists the visibility of scientific research is manifold that lead to increase in research output Ex. Physics Archival Digital library
- 3. Preserving traditional culture: Digital libraries preserve indigenous culture of the tradition, language, anthology and traditional music etc. in audio and video formats. These initiatives made endangered documents digitally available.
- 4. Exploring popular music: DLs archive different music of the various generations. The digital libraries created with scientific information like e-journals, databases.

- Gorman's laws are the most famous. He has reinterpreted Ranganathan's laws in the context of today's library and its likely future. Michael Gorman has given us his five new laws of librarianship:
- *Libraries serve humanity.*
- *Respect all forms by which knowledge is communicated.*
- *Use technology intelligently to enhance service.*
- *Protect free access to knowledge; and*
- *Honor the past and create the future* (Crawford & Gorman, 1995).

- **The Five Laws of the Web**
- The *Five Laws of the Web* are inspired by the "*Five Laws of Library Science*" which were the seed of all of Ranganathan's practice. These laws form the foundation for the Web by defining its minimum requirements. While the laws seem simple on first reading, think about some of the conversations on the Web and how neatly these laws summarize much of what the Web community believes. Although they are simply stated, the laws are nevertheless deep and flexible. These laws are:
- *Web resources are for use.*
- *Every user his or her web resource.*
- *Every web resource its user.*
- *Save the time of the user.*
- *The Web is a growing organism*

Digital Library Vision

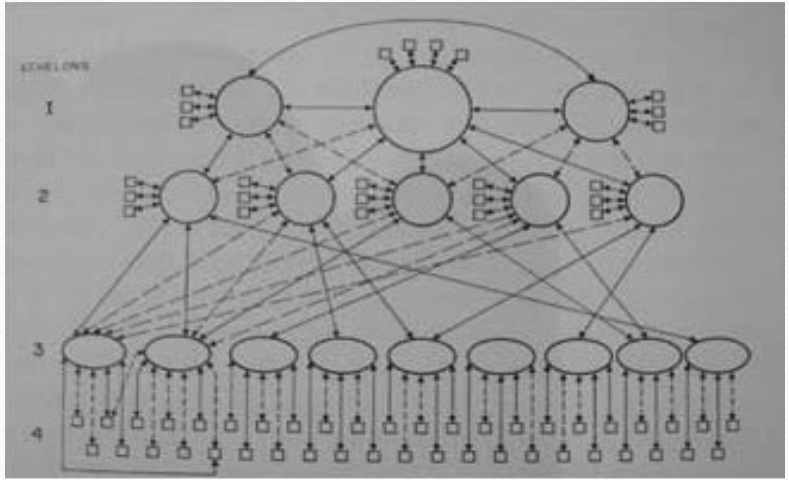
V. Bush 1945



MEMEX in the form of a desk would instantly bring files and material on any subject to the operator's fingertips. Slanting translucent viewing screens magnify supermicro-film filed by code numbers. At left is a mechanism which automatically photographs longhand notes, pictures and letters, then files them in the desk for future reference.

AS WE MAY THINK CONTINUED

J. Licklieder 1965



Digital Library Definition

Systems that support searching, use, creation of content

Institutions with people, digital collections, and services

Repositories of digital data and documents, as a component of cyber-infrastructure, e-research, e-science, e-social science, e-learning... institutional repositories, open archives, data collections

Borgman, 2005

Digital libraries are sociotechnical systems—networks of technology, information, documents, people, and practices

Ann Bishop et al. 2003

Digital libraries are moving beyond personal intellectual protheses to become much more participative and reflective of social history.

Marchionini Gary 2012

- Digital Library Initiative (Fox, 1993) Summarizing from workshops conducted in 1991 and 1992, this definition states that a National Electronic Library is (1) a service; (2) an architecture; (3) a set of information resources, databases of text, numbers, graphics, sound, video, etc.; and (4) a set of tools and capabilities to locate, retrieve, and utilize the information resources available. The users of a national electronic library would include students, teachers/professors, researchers/scholars, librarians, authors, publishers, information providers, and practitioners. Contributors of information resources would include publishers, universities, professional societies, libraries, authors, editors, and compilers.

- The above definition remains among the most comprehensive by including services, architecture, content, enabling technologies, users, and content. It provided a basis for further discussion and refinement. The Digital Library Initiative, announced in September, 1993 (and since dubbed ÒDLI-1Ó), defined the term only implicitly (National Science Foundation, 1993), stating that Borgman, IPM DL paper, final revised version, October 23, 1998, page 13 ÒInformation sources accessed via the Internet are the ingredients of a digital library

Digital libraries are a set of electronic resources and associated technical capabilities for creating, searching, and using information. In this sense they are an extension and enhancement of information storage and retrieval systems that manipulate digital data in any medium (text, images, sounds; static or dynamic images) and exist in distributed networks. The content of digital libraries includes data, metadata that describe various aspects of the data (e.g., representation, creator, owner, reproduction rights), and metadata that consist of links or relationships to other data or metadata, whether internal or external to the digital library.

- Defining Elements of Digital Libraries Borgman, IPM DL paper, final revised version, October 23, 1998, page 16 Several aspects of these definitions should be noted. One is that digital libraries are viewed as databases, albeit databases of rich content, whether full text, images, or combinations of media and representations. Much digital library research, particularly that conducted in departments of computer science, focuses on Òenabling technologiesÓ such as database structure, retrieval algorithms, filtering, intelligent agents, network architecture, and other necessary capabilities.

- Digital Libraries as Institutions or Services Borgman, IPM DL paper, final revised version, October 23, 1998, page 17 The terms digital library, Òelectronic library, and virtual library have appeared in the professional literature of library and information science for some years already, but rarely with explicit definitions. Lyman (1996), in an article entitledWhat is a digital library? Technology, intellectual property, and the public interest, explores concepts he views to be prerequisite to defining the concept of a digital library, such as electronic publishing and digital documents. Young (1996) lists characteristics of digital libraries, saying they provide personalized or custom services for accessing, assembling, and analyzing information resources from a variety of diverse sources in many different formats (1996, p 122).

- Given the rapid expansion of computer networks, distributed access to information resources, electronic publishing, distance-independent learning, electronic commerce, and related technologies, vastly more research on all aspects of digital libraries is needed. Technological developments, structural changes in the way universities are funded, escalating costs of information resources, demographics, and other factors combine to make the transformation of research libraries an urgent challenge. Attention to digital libraries research and practice, under all of these definitions, will continue to increase for the foreseeable future.

- The goals of DLI-1 were modest by todayÕs standards. Research was supported in three areas: (1) capturing data and metadata of all forms (text, images, sound, speech, etc.) and categorizing and organizing them, (2) advanced software and algorithms for browsing, searching, filtering, abstracting, summarizing and combining large volumes of data, imagery, and all kinds of information; and (3) the utilization of networked databases distributed around the nation and around the world.

- Research-oriented Definitions One of the first meetings to focus directly on digital libraries issues was a 1991 workshop on Future Directions in Text Analysis, Retrieval and Understanding (summarized in Fox, 1993) and a white paper on electronic libraries that followed from it (Lesk, Fox, & McGill, 1991). The resulting excitement led to more workshops that refined a research agenda and eventually led to the digital libraries initiatives.

DL-I

- The major six projects undertaken in Phase-I Digital Library Initiatives from 1994 and ran for four years i.e. 1998 are
- • University of California at Berkeley — Environmental Planning Library and Geographic Information Systems. http://elib.cs.berkeley.edu
- • University of California at Santa Barbara – Alexandria Digital Library project; spatially referenced Map Information. http://alexandra.ucsb.edu
- • Carnegie Mellon University Project – Informedia Digital Video Library http://www.infomedia.cs.cmu.edu
- • University of Illinois at Urbana-Champaign – Federating Repositories of Scientific Literature. http://dli.grainger.uiuc.edu
- • University of Michigan Digital Library Project (UMDL): Intelligent Agents for Information Location. http://www.si.umich.edu/UMDL
- • Stanford University - Interoperation Mechanisms among Heterogeneous Services. http://www.diglib.stanford.edu Phase II(www.dli2.nsf.gov) digital library initiative

DL-II

- www.dli2.nsf.gov
- DLI-2 is an interagency program sponsored by
- National Science Foundation (NSF)
- Defense Advanced Research Projects Agency
- National Library of Medicine
- Library of Congress
- National Endowment for the Humanities
- National Aeronautics & Space Administration
- Federal Bureau of Investigation
- The program operates in partnership with
- Institute of Museum and Library Services
- Smithsonian Institution
- National Archives and Records Administration

- Within the NSF, the initiative receives support from the Directorates for Computer and Information Science and Engineering; Social, Behavioral and Economic Sciences; and Education and Human Resources.
- NSF serves as the administrative agent for the initiative's competitions and funded projects. Policy, planning and programmatic decisions are made by an interagency management group in which representatives of each sponsoring agency and the NSF directorate participate.

E-learning Vision

Pask: Saki 1950

Illinois: Plato 1960

ITS 1980

Post Web

VLE, LMS

2004 SCORM Reusable learning object

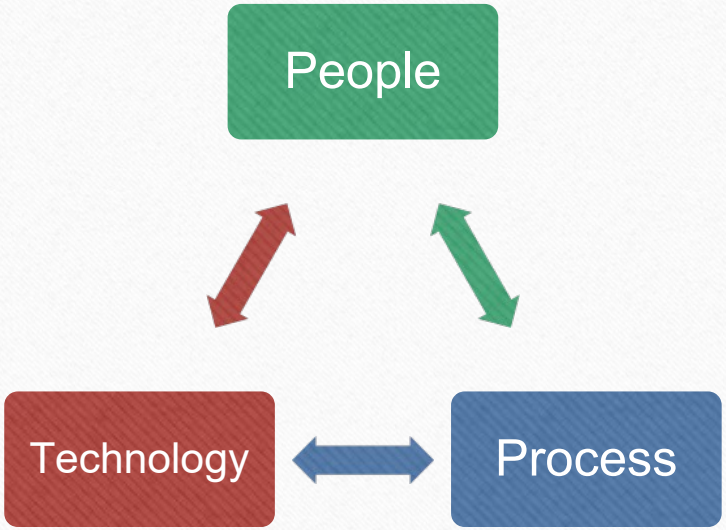
E-learning inspires individual/community engagement and inclusive **learning** environments through the use of emerging technologies

E-learning definition

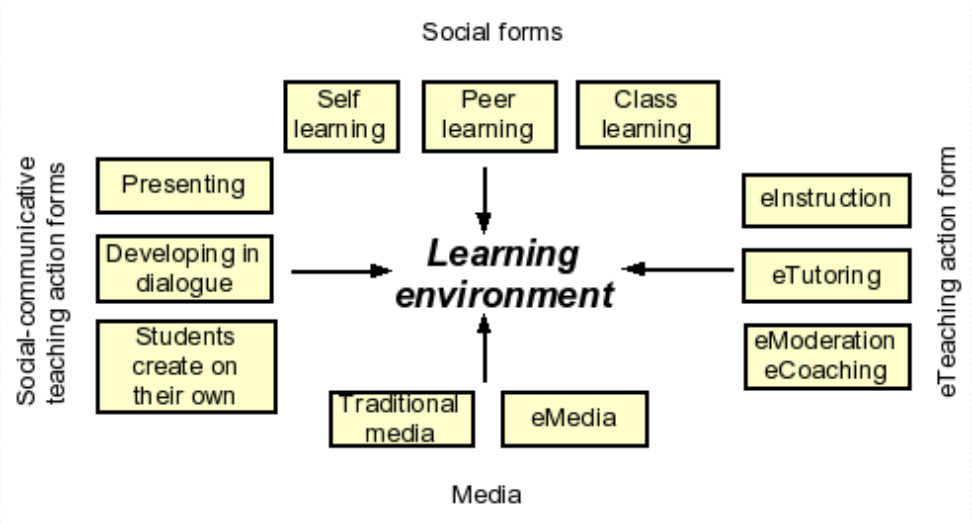
Learner uses technology to interact with instructor and other learners and some form of support is given to learners

Other: Computer assisted instruction (CAI), computer assisted learning (CAL), computer-based education (CBE), distance learning, online learning, learning management systems (LMS), self-directed learning (SDL), and massive open online courses (MOOC).

Digital Library and E-learning Theoretical framework



Learning environment



Euler, D. Didaktische Gestaltung von E-learning-unterstützten Lernumgebungen 2004

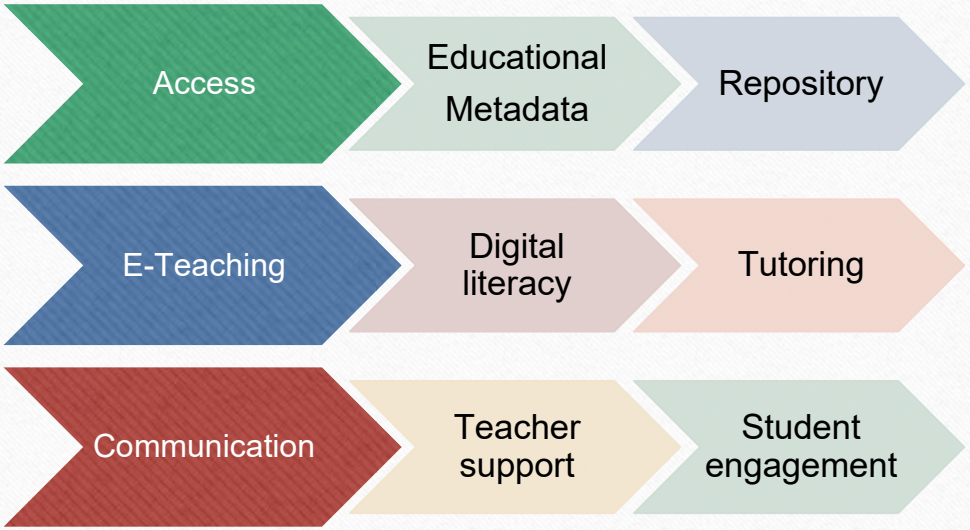
E-learning social interaction

Self-learning

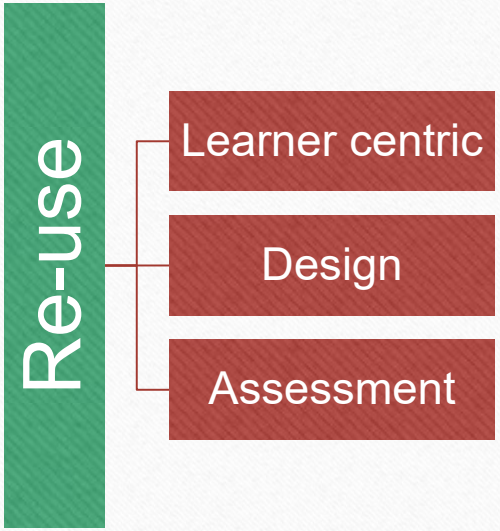
Peer learning

Classroom learning

Digital library as Learning Environment



Learning Object



NSDL

National Science Digital Library

OER Collection

NSDL has been investigating ways to enable the use and repurposing of educational resources to support student-centered differentiated instruction.

Learning Application Readiness: readiness of the collection in the path of users such as LMS, registries, curriculum

Curriculum Customization Service: a pilot study to help teachers and their school to find, save, organize, share and reuse learning objects

OER educational metadata

NSDL uses frameworks that characterize resources for identification, discovery, use:

1. subject,
2. education level,
3. resource type,
4. audience,
5. language,
6. copyr. and
7. educational standards.

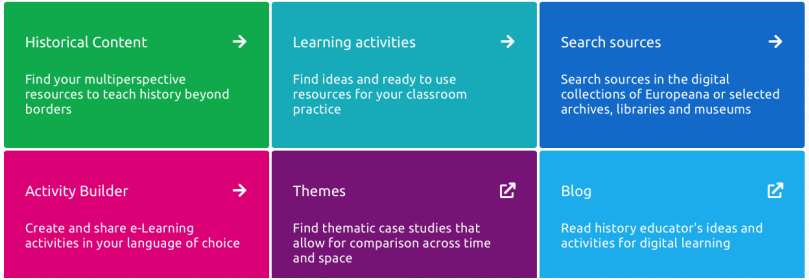
Europeana

European Digital Library

Organization and Policy

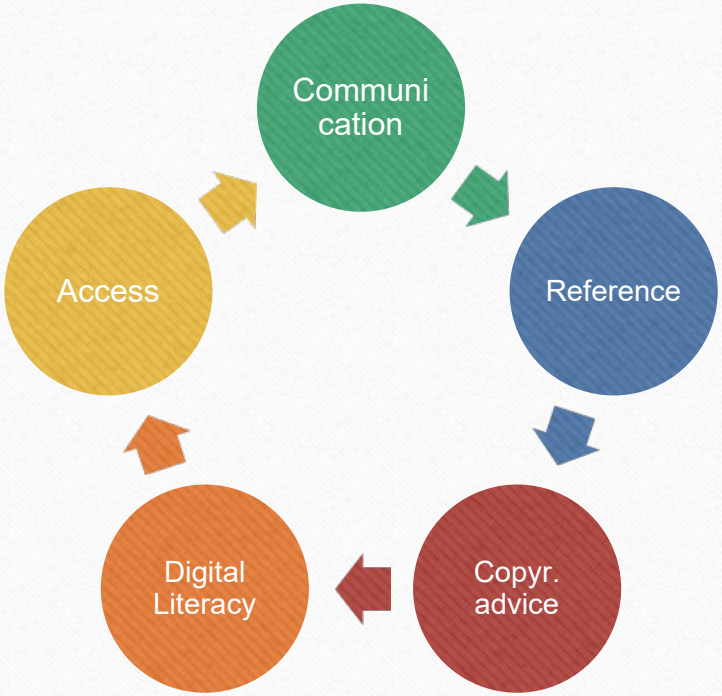
- Europeana for Education and Learning
- The Learning Resource Exchange
- Project Historiana Learning

Historiana offers free historical content, ready to use learning activities, and innovative digital tools made by and for history educators across Europe.



Digital librarians as educators

- Competencies



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Thank you of your attention!